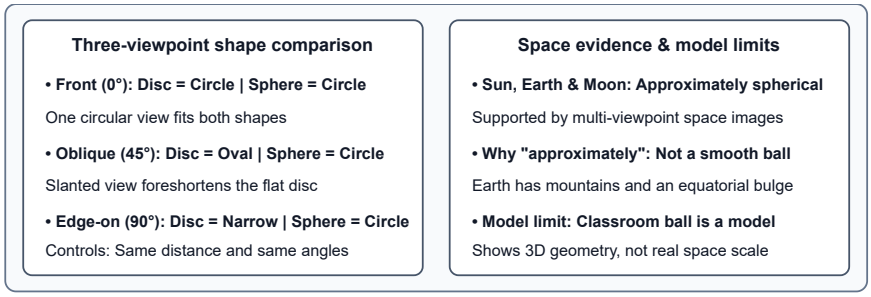
Knowledge organiser



Knowledge organiser model. It compares a flat disc and a sphere at 0°, 45°, and 90° and links space evidence.

Learning goal. I can compare a disc and a sphere across viewpoints and explain spherical evidence.

Tier 2 and Tier 3 vocabulary

Tier.	Word.	Meaning.	Example in this lesson.
Tier 3	sphere	A three-dimensional round solid that keeps a circular outline from every viewpoint.	A sphere keeps a circular outline at 0°, 45°, and 90°.
Tier 3	disc	A flat circular shape that looks narrow when turned edge-on.	A flat disc looks circular from the front but narrow at 90°.
Tier 2	viewpoint	The angle and position from which an observer looks at an object.	Changing the viewpoint from 0° to 90° separates a disc from a sphere.
Tier 2	evidence	Observations from multiple views or spacecraft that support a science claim.	Multiple spacecraft viewpoints provide strong evidence for Earth's round shape.
Tier 2	approximately	Very close to a spherical shape, though slightly flattened with surface features.	Earth, the Moon, and the Sun are approximately spherical bodies.
Tier 2	limit	What a classroom ball model cannot show, such as real planetary scale.	One limit of a smooth ball model is omitting mountains and Earth's bulge.

Core Key Stage 3 science ideas. A single circular outline at **0°** cannot distinguish a flat **disc** from a three-dimensional **sphere**. Keep viewing distance equal across **viewpoints** (**0°**, **45°**, and **90°**). The disc becomes oval and then narrow, whereas the sphere stays circular. Multi-viewpoint

spacecraft **evidence** shows the Sun, Earth, and Moon are **approximately** spherical. A classroom ball has honest **limits** (it omits real scale, surface relief, and Earth's bulge).

Made by Matt